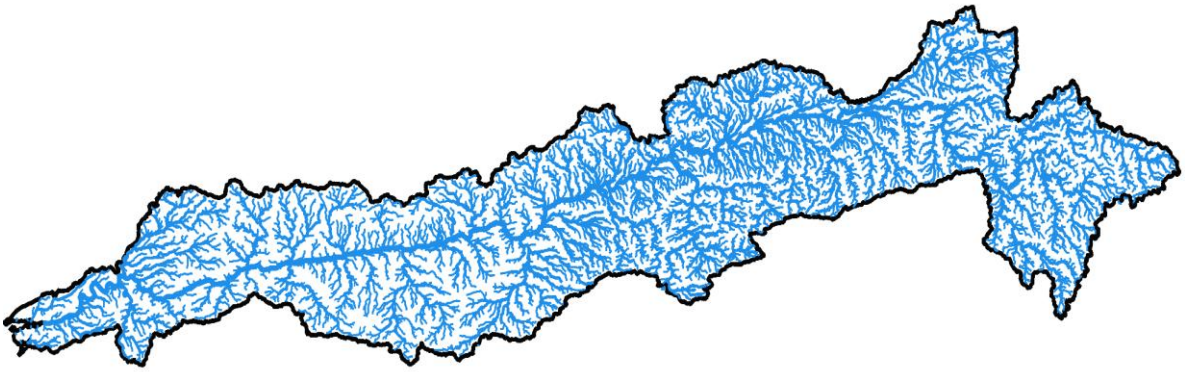
National River Conservation Directorate
Ministry of Jal Shakti, Department of Water Resources,
River Development & Ganga Rejuvenation
Government of India

Mapping of Best Practices in Solid-Liquid Waste Management - International and National



March 2026

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National River Conservation Directorate (NRCD)

The National River Conservation Directorate, functioning under the Department of Water Resources, River Development & Ganga Rejuvenation, and the Ministry of Jal Shakti, provides financial assistance to the State Government for the conservation of rivers under the Centrally Sponsored Schemes of 'National River Conservation Plan (NRCP)'. National River Conservation Plan to the State Governments/ local bodies to set up infrastructure for pollution abatement of rivers in identified polluted river stretches based on proposals received from the State Governments/ local bodies. (www.nrcd.nic.in)

Centres for Narmada River Basin Management and Studies (cNarmada)

The Center for Narmada River Basin Management and Studies (cNarmada) is a Brain Trust dedicated to River Science and River Basin Management. Established in 2024 by IIT Gandhinagar and IIT Indore, under the supervision of cGanga at IIT Kanpur, the center serves as a knowledge wing of the National River Conservation Directorate (NRCD). cNarmada is committed to restoring and conserving the Narmada River and its resources through the collation of information and knowledge, research and development, planning, monitoring, education, advocacy, and stakeholder engagement. (www.cnarmada.org)

Centres for Ganga River Basin Management and Studies (cGanga)

cGanga is a think tank formed under the aegis of NMCG, and one of its stated objectives is to make India a world leader in river and water science. The Centre is headquartered at IIT Kanpur and has representation from most leading science and technological institutes of the country. cGanga's mandate is to serve as think-tank in implementation and dynamic evolution of Ganga River Basin Management Plan (GRBMP) prepared by the Consortium of 7 IITs. In addition to this, it is also responsible for introducing new technologies, innovations, and solutions into India. (www.cganga.org)

Acknowledgment

This report is a comprehensive outcome of the project jointly executed by IIT Gandhinagar (Lead Institute) and IIT Indore (Fellow Institute) under the supervision of cGanga at IIT Kanpur. It was submitted to the National River Conservation Directorate (NRCD) in 2026. We gratefully acknowledge the individuals who provided information for this report.

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PREFACE

The Narmada River, often referred to as the lifeline of central and western India, holds unparalleled significance for the region's water resources, ecosystems, and communities. Its hydrological complexity, combined with its ecological and cultural heritage, underscores the critical need for its preservation and sustainable management.

This report, focusing on case studies of national and international waste management systems, aims to provide an in-depth analysis of the plans and technologies used in different countries for solid-liquid waste management. The collection and compilation of data on solid-liquid waste management strategies will support the development of plans and strategies for waste management in the Narmada River basin.

The findings and recommendations presented in this report are intended to serve as a guiding framework for policymakers, researchers, and practitioners committed to sustainable water management in the Narmada Basin. By identifying trends and challenges, it aspires to support the development of policies that balance ecological preservation with socioeconomic progress.

We are deeply grateful to all individuals, organizations, and institutions that contributed to the preparation of this report. Their dedication, expertise, and support have been instrumental in shaping this study. It is our hope that this report will inspire collaborative efforts and meaningful action toward the long-term health and sustainability of the Narmada River and its Basin.

Centre for Narmada River Basin Management and Studies (cNarmada)

IIT Gandhinagar, IIT Indore

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Abbreviations

C&D	Construction and Demolition
CTECH	Cyclic Activated Sludge Technology
EPA	Environmental Protection Agency
EPM	Empresas Públicas de Medellín
EU	European Union
MBBR	Moving Bed Biofilm Reactor
MBR	Membrane Bioreactor Technology
MLD	Million Liters per Day
MSW	Municipal Solid Waste
RDF	Refuse-derived fuel
SBR	Sequential Batch Reactor
SDG	Sustainable Development Goal
SLWM	Solid and liquid waste management
STP	Sewage Treatment Plant
TPD	tons per day
UASB	Upflow Anaerobic Sludge Blanket
USEPA	United States Environmental Protection Agency
WHO	World Health Organization
WTE	Waste-to-Energy
WtF	Waste-to-Energy Facility

1. Overview of the Narmada Basin

The Narmada River Basin extends from 21°40'12" to 23°41'24" N latitudes and 72°48'36" to 81°45'36" E longitudes, encompassing a total area of 98,796 sq. km, which constitutes about 3% of India's total geographical area (3,297,427.32 sq. km). Bounded by the Vindhyan ranges to the north, the Maikala hills to the east, the Satpura ranges to the south, and the Arabian Sea to the west, the basin lies at the northern extremity of the Deccan Plateau and covers significant portions of Madhya Pradesh and Gujarat, with smaller parts extending into Chhattisgarh and Maharashtra. The basin spans 40 districts, comprising 27 districts of Madhya Pradesh, 07 districts of Gujarat, 04 districts of Chhattisgarh, and 02 districts of Maharashtra. The Narmada Basin exhibits an elongated shape, stretching about 915.65 km east to west and about 236 km north to south. The basin experiences a tropical monsoon climate characterized by distinct wet and dry seasons. The Narmada River, the longest west-flowing river in India, drains into the Arabian Sea near Bharuch, Gujarat, and is further subdivided into the Upper, Middle, and Lower Narmada River Basins(cNarmada River Atlas,2025).

Data Sources: For this report, data have been collected from various official international government reports focused on best practices for solid-liquid management. This report provides policymakers and planners with valuable resources to inform their decisions.

2. Introduction

Waste is any substance or object discarded as a result of human activities. It exists in multiple forms, including solid waste, municipal solid waste (MSW), electronic waste (e-waste), construction and demolition waste, hazardous waste, and liquid waste such as sewage, greywater, and industrial effluents. The generation of waste is an inevitable consequence of economic development, urbanization, and changing consumption patterns, making its effective management a critical global concern.

Globally, waste generation has reached unprecedented levels. Billions of tonnes of municipal solid waste are produced annually, with projections indicating that this will rise to nearly 4 billion tonnes by 2050. The magnitude of this challenge is such that, if compacted into shipping containers, global waste could circle the Earth's equator multiple times. In addition, approximately 54 million tonnes of e-waste were generated worldwide in 2019, and this figure is expected to increase to 75 million tonnes by 2030. Beyond municipal waste, significant quantities of agricultural, industrial, construction, domestic,

and healthcare waste are generated by diverse sources, including farms, factories, construction sites, and healthcare facilities, further intensifying environmental pressures (United Nations Environment Programme and International Solid Waste Association, 2024).

In the Indian context, the challenge is particularly acute. Rapid urbanization, population growth, and industrial expansion have generated over 62 million tonnes of solid waste annually, along with substantial volumes of liquid waste; in many urban areas, more than 70% of sewage remains untreated. Inefficient waste management systems contribute to serious environmental degradation, including water pollution, soil contamination, greenhouse gas emissions, and adverse impacts on public health. Government initiatives such as Swachh Bharat Mission (SBM) 2.0 and Namami Gange have significantly improved waste management practices; however, critical gaps persist in terms of infrastructure, technological integration, decentralized treatment systems, and effective implementation.

Against this backdrop, there is an urgent need to identify, assess, and replicate successful waste management practices. This report maps best practices in solid and liquid waste management at the national and international levels, drawing on Indian case studies such as Madhya Pradesh, Gujarat, and Kerala, as well as global examples. It highlights innovative and scalable approaches, including GIS-based monitoring, machine learning for predictive analysis, decentralized treatment systems, circular economy principles, and community-led bioremediation. By synthesizing these practices, the report aims to provide policymakers, researchers, and practitioners with practical insights to promote sustainable, resilient, technology-driven waste management systems aligned with global goals and India's net-zero target.

In addition to municipal waste, various types of waste produced by human activities in considerable amounts include agricultural, construction and demolition, industrial, domestic, and healthcare waste, and their sources are farms, construction sites, factories, and hospitals, as shown in Figure 1 (United Nations Environment Programme,2024, United Nations Environment Programme, & United Nations Human Settlements Programme. 2021).

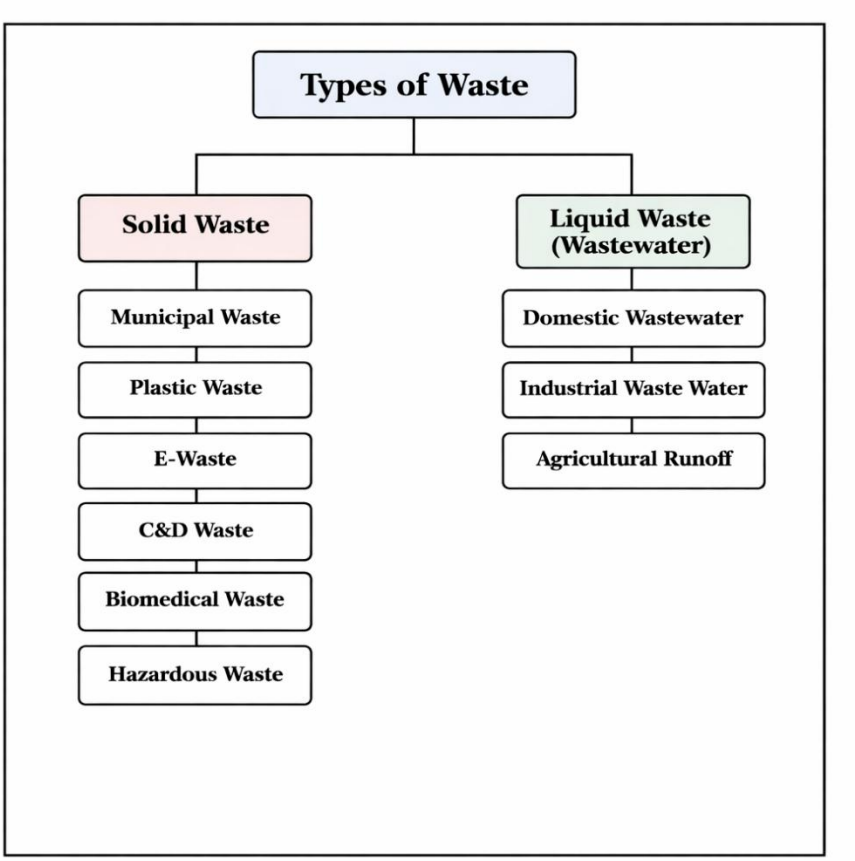


Figure 1: Types of solid and liquid waste

The generation of municipal waste depends entirely on humans: how they purchase, their daily practices, and how they manage waste at home. Climate change, pollution, and biodiversity loss are all linked to municipal waste generation and management. Improper municipal waste management and disposal contribute to greenhouse gas emissions, which ultimately lead to climate change. Similarly, there are different types of pollutants, such as plastic waste and toxic chemicals, that affect ecosystems and biodiversity. There is a significant diversity in the waste, which can be categorised in different ways. One method of classification is based on the material, such as plastic or food waste, while another is based on the product type, such as e-waste. Apart from this, another way is to categorize waste by the product type that generates it, e.g., municipal solid waste, industrial waste, etc. Figure 2 shows the categories of waste (United Nations Environment Programme,2024).

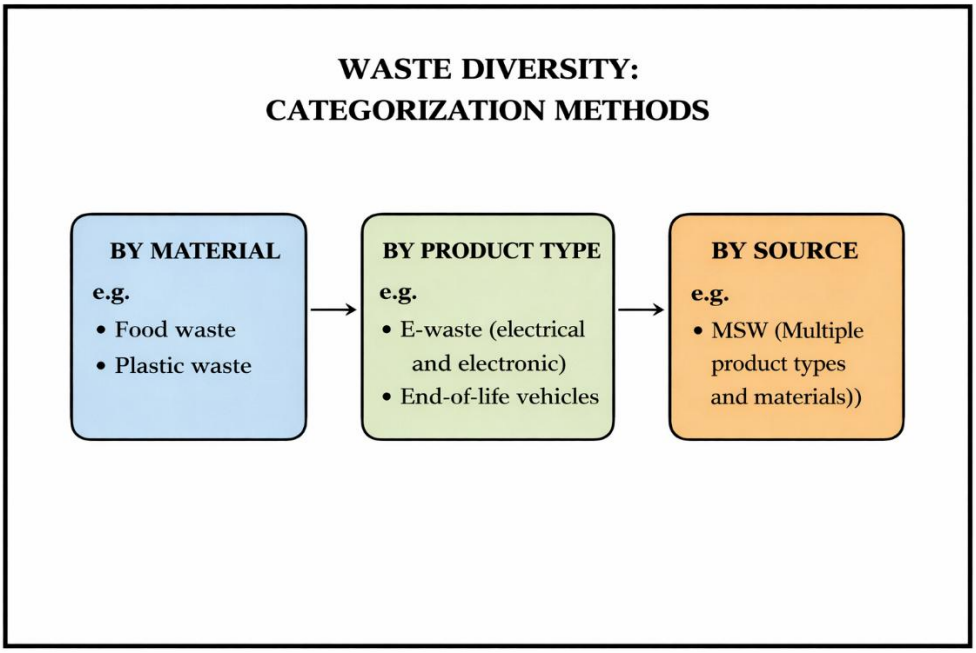


Figure 2: Waste diversity Categorization on the basis of material, product, and source of generation.

Waste management, both solid and liquid, is a crucial global task for a better ecosystem, supporting environmental health, public health, and sustainable development. Improper management of waste can lead to various health issues and affect the aquatic life through contaminated water, soil, and air. These improper waste management practices create hazardous conditions for workers in the management sector and for local communities near waste sites.

Impact of Poor Waste Management

There are many adverse effects of poor waste management, whether it is solid or liquid waste; both can contribute equally to environmental pollution on land and in water bodies such as rivers, lakes, and oceans.

2.1. Human Health/Disease Transmission:

Blockages in drains caused by solid waste can lead to flooding and stagnant water, which, in turn, can spread vector-borne diseases. Similarly, if there is no separation of urine and faecal matter from solid waste, there is a high risk of spreading diseases such as typhoid and cholera. In addition, mismanaged solid waste and open dumpsites contaminate surface and groundwater, which, in turn, leads to waterborne disease, as these are the primary sources of drinking water. Improper and open burning leads to air pollution, which can be

hazardous to human health, and residents living near these dumpsites experience more severe health effects.

2.2. Ecological Disturbance:

The introduction of pollutants, such as plastic, heavy metals, and chemicals from industrial discharges, into water bodies ultimately enters the food chain of the organisms in those bodies, affecting both aquatic organisms and humans who consume them, posing health risks. Additionally, if the dumpsite is not managed properly, leachate from the dumpsite contaminates soil and water bodies, harming the surrounding ecosystem. Burning waste generates greenhouse gases like methane, which contribute to climate change, and open burning also affects air quality and, in turn, forms air pollutants.

2.3. Socioeconomic decline:

There is an urgent need for proper waste management practices, as improper management leads to negative socio-economic effects that affect living standards, community relations, and overall economic growth in that place. If cities and beaches are clean, there will be more tourism and higher property values, which in turn will boost the local economy. Programs that reduce waste can, in turn, lower collection and transport costs, thereby reducing fuel costs. Apart from this, proper waste management will have a positive effect on the high-risk group, as there will be less trash, leading to fewer respiratory and skin diseases and, ultimately, reducing the costs on medical facilities.

The waste hierarchy is a way of deciding how to deal with waste in the best order. The first and most important step is preventing waste. If waste cannot be prevented, the next best options are reusing, recycling, and recovering materials. However, no matter how effective these steps are, some waste will always remain, so safe disposal facilities remain necessary to protect people's health and the environment.

Solid-liquid waste management is a complex process that involves many interconnected steps, including reducing waste generation, collecting waste, transporting it, recycling, recovering useful materials, and disposing of what remains. As all the steps are interconnected, it is very important to carefully plan, monitor, and manage these procedures. There is no chance of mishandling; otherwise, it leads to health problems, such as a disease outbreak. Waste management is a challenging task due to rapid urbanization and high population growth. Lack of space, weak laws, policies, and budget constraints make the situation worse(EPA,2020). Common challenges in solid-liquid waste

management are listed in Table 1. There is a need for a robust system, sound planning, and effective leadership to support waste management practices that, in turn, protect health and the environment.

Table 1: Common Challenges in Solid-Liquid Waste Management (EPA,2020)

Category	Specific Challenges	Impact and Details
Financial	Limited financial capacity	Many cities lack sufficient budgets or long-term investment for effective waste management. As they are responsible for proper management, but due to rising costs, population growth, and limited expertise, they are constrained in their efficiency.
Institutional	Political turnover and lack of coordination	As the government changes, the higher staff also change, which either stops ongoing projects or requires staff changes. This staff change affects long-term projects such as solid waste management and reduces their efficiency. Apart from this, all sections of the SWM fall under the ministry and local bodies, and there should be proper coordination among them for efficient management.
Technical	Incompatible equipment and expertise	Many cities import SWM equipment, which can lead to high maintenance expenses, especially when local people lack technical expertise. Moreover, many governments are not technically sound in waste management best practices and outsource to private companies, which, in turn, set specific requirements that, if not met, cause the projects to stop.
Operational	Lack of Staff and difficult working conditions	There are staff for waste emergencies, but not for long-term planning and development of waste management. In many cities, staff members are underpaid and undertrained, which increases the risk of disease.
Environmental	Geography, topography, and climate	Different climate conditions affect the costs, equipment requirements, and technologies. The amount of waste produced depends on cultural habits and increased consumption.
Social	Cultural growth and lack of communication	As financial conditions improve, consumption of goods increases, leading to greater waste. Other than this lack of knowledge among the population regarding illegal dumping, segregation, and not paying fees, leads to complications.
Physical	Land limitation	As the population grows, waste generation increases, making it difficult to manage. As less space will be available for solid waste management facilities.

3. Solid- Liquid Waste Management Solid- Liquid Waste Management in different countries

Global Trends in Solid-Liquid Waste Management describe patterns, changes, and challenges worldwide. In this journey of waste from generation to disposal, including collection, treatment, and reuse, various international government reports highlight the increasing amounts of waste generated due to urbanization, population growth, economic development, and changes in lifestyle. In this section, the focus is on analysis and learning from various case studies for solid-liquid waste management.

4. Ireland Case Study

(IEA Bioenergy, March 2025)

4.1. Waste Framework Directive (EU) 2008

In 2008, the Waste Framework Directive was incorporated into European Union law. This legislation aims to establish common rules for waste management across all EU member states, ensuring that they adhere to a unified framework.

The directive primarily focuses on minimizing waste production and enhancing recovery targets, leading to several key outcomes:

- A gradual reduction in the overconsumption of natural resources
- A decrease in the loss of biodiversity and habitat destruction
- A reduction in greenhouse gas emissions

Within the Waste Framework Directive, various types of waste are governed by specific regulations, such as those on the management of sewage sludge in agriculture and the landfill directive.

4.2. Waste Hierarchy Concept

The basis of EU waste management is the Waste Hierarchy established by the Waste Framework Directive, which places landfill disposal as the last option and prevention as the best. The Waste Hierarchy is shown in Figure 3.

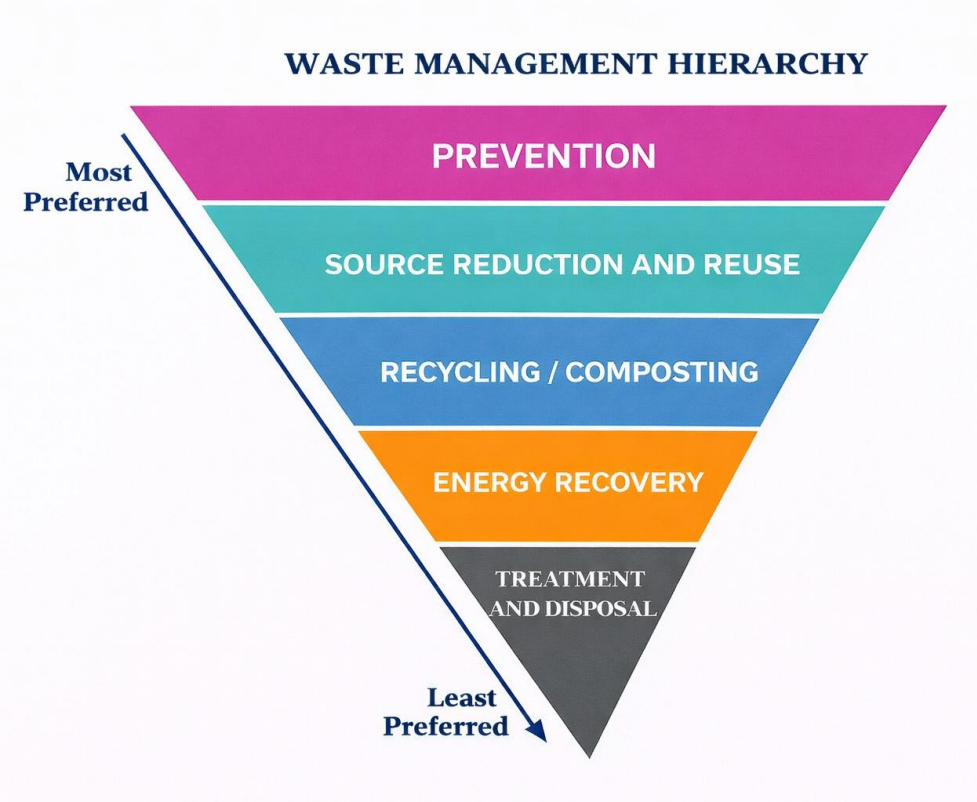


Figure 3: Waste management hierarchy illustrating the most preferred and the least preferred methods.

Later in 2019, the EU launched the Green Deal, a roadmap to make the EU climate-neutral by 2050 and to develop a resource-efficient, competitive economy. One of the major parts of this green deal is the Circular Economy Action Plan, which aims to strengthen the EU economy and protect the environment. They focus on product production and provide detailed information about the eco-friendly product to help consumers buy it and get detailed information about the products they buy. They also believe in using resources more sustainably and recycling.

The EU sets clear targets by 2025: to recycle more than 55% of municipal waste and 65% of packaging waste, and to reduce landfill rates.

In 1998 and 2002, Ireland followed EU policies and introduced its own Changing Our Ways Policy (Environmental Protection Agency, 1998) to reduce waste generation, reduce dependence on landfills, and accept incineration as a waste-to-energy alternative.

In 2002, the Preventing and Recycling Waste strategy was launched, adopting the Waste hierarchy, which is quite similar to the EU system. In this, the first point is prevention, reuse, recycling, and biological treatment are part of it; then incineration; and, lastly,

disposal (Landfilling). More policies later supported this approach, and as a result, Ireland gradually shifted to a waste-to-energy sustainable solution from landfill-based systems.

Ireland implemented a Waste Action Plan for the Circular Economy, a National Policy focused on the preservation of natural resources rather than waste disposal, covering the years 2020-2035, and aligned with the EU Green Deal. There are specific targets for different types of waste, which include household, industrial, C&D, plastics, and textiles.

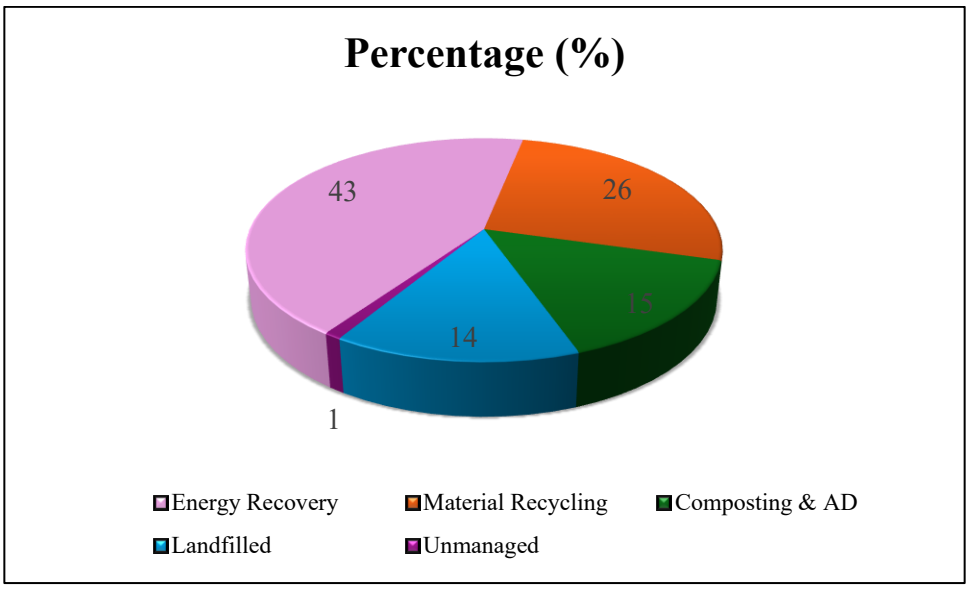


Figure 4: Municipal Treatment in Ireland (2022)

4.3. Ireland: Meath Waste-to-Energy Facility

This grate incinerator is run by Indaver Ireland Ltd., located near Duleek in County Meath. It is the first of its kind and began operations in August 2011.

Treatment Capacity: 235,000 tonnes Municipal Waste /Year

Generates: 18MW of electricity

Total Material and Energy Recovery rate: 98.9% (Non-Hazardous) &100 % (Hazardous)

Waste streams treated are given in Table 2.

Table 2: Waste streams treated at the Meath Waste-to-Energy Facility

Waste category	Description
Residual Waste	Non-recyclable waste from domestic & commercial/industrial kerbside collection
Combustible residual waste Reject Fractions	Combustible residual waste from recycling processes
Refuse-derived fuel (RDF)	A processed fuel made from combustible parts of municipal or industrial waste
Skip Waste	Residual non-bulky combustible
Contaminated Materials	Packaging/clothing / PPE / liners
Industrial discharge	wash waters from the pharmaceutical industry
Sludge	Wastewater treatment sludge

Technologically, in a waste-to-energy facility, waste is accepted and incinerated in a moving incinerator. The detailed process is shown in Figure 5. During this process, steam is generated for electricity production, which is used on-site and exported. Ferrous and non-ferrous metals are recovered from the non-hazardous ash left after waste burning, and the ash and flue gas are also pretreated to protect the environment. After combustion, the boiler ash and flue gas are sent for recovery to a salt mine in Northern Ireland.

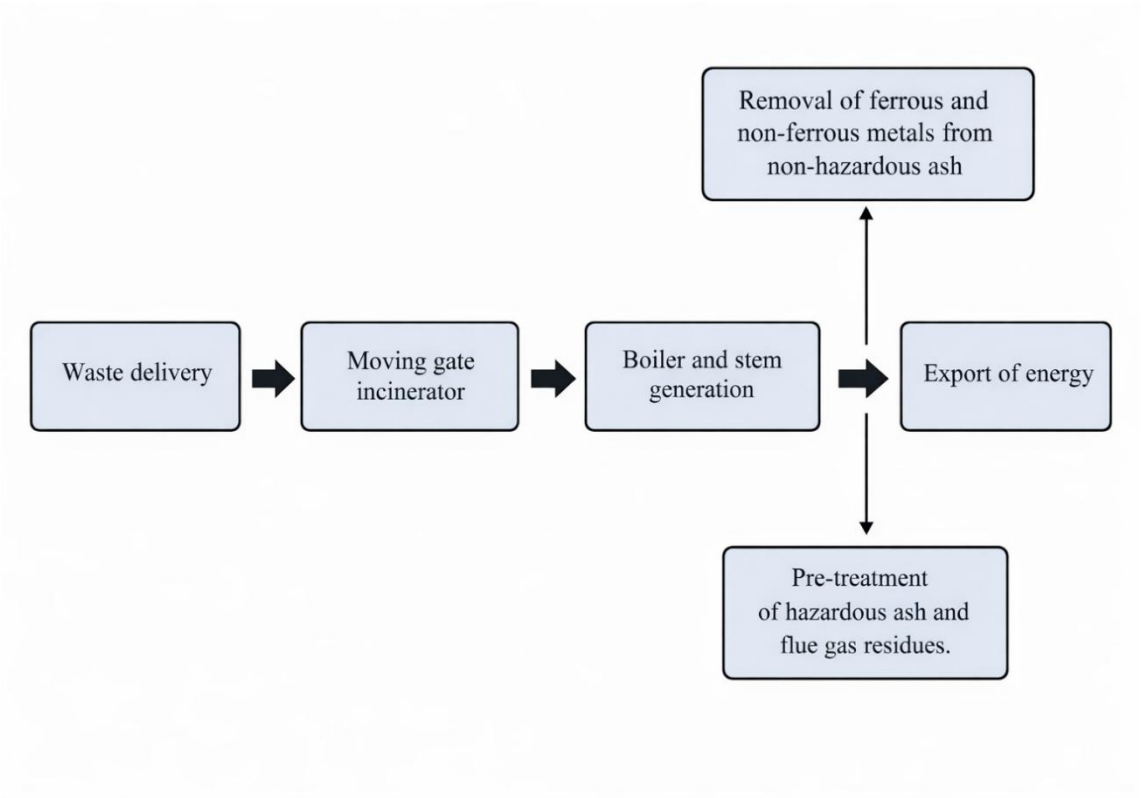


Figure 5: Technology description of Waste-to-Energy facility in Ireland

Indaver's approaches to improve the sustainability of the Waste-to-Energy facility in Ireland are listed in Table 3.

Table 3: Different types of Categories and Initiatives

Category	**Initiative**
Renewable Energy	Exploring the option of storing the plant's renewable energy in batteries that can be used when wind or solar sources are not generating.
Low-emission Transport	Providing e-delivering trucks for all on-site transport, with on-site charging infrastructure.
Biodiversity	The WtE plant has laying lawns, flower meadows, bushes, and hedges, which reduce the use of pesticides.
Ecological Support	The site has ponds and beehives.
Site Design	The facility's horizon matches the landscape's curves.

5. US Case Study

Source: (IEA Bioenergy,2025)

Green Era Renewable Energy & Urban Agriculture in Illinois

The Green Era Renewable Energy & Urban Agriculture campus is a community center for renewable energy generation, urban programming, and education. This campus is situated on a 9-acre former contaminated site in the Aurburn-Gresham neighbourhood of Chicago, Illinois. To complete this green Era campus, 10 years of grassroots community organising were required.

It's a collaborative project involving multiple partners to fulfill the project requirements.

The main technology of the project is an Anaerobic digester, whose construction started in 2020 and was completed in 2022. The Green Era Campus was officially launched in 2024.

The main component of the Anaerobic digester process is approximately 80,000 tons of food/day. The sources of food collection are restaurants, food-based industries, and households. This system is an alternative to the existing facility of landfilling food.

In this facility, various types of food waste are processed, as listed in Table 4.

Table 4: Different types of waste and processing

Waste Type	Source
Solid food waste	Food processors, grocery stores, and restaurants
Household food waste	separated bagged kitchen waste from neighbourhood curbside collection programs.
Liquid waste	Wash water, whey, liquid ice-creams mix, soups, sauces& syrups.
Fats, oils, and grease	Grease traps from restaurants and food processing facilities

Step-by-step process of technology

- Collection of food waste from commercial and residential sources and delivery to the facility.
- Both solid and liquid waste are processed inside the building into a homogenized slurry after screening.
- A separation mill pulps the organic material and removes non-digestible contaminants such as plastic.

- After this pulped material passes through a grit separation system, where heavy materials such as stones, sand, glass, etc, are removed.
- Then this cleaned and conditioned feedstock is stored in a holding tank before processing in the digester.
- After transfer into the digester, the inside temperature is 130°F in anaerobic conditions, where breakdown of organic material occurs and produces biogas.
- The composition of biogas is 65% methane and 35% carbon dioxide. This biogas is sent to an upgraded instrument for further processing, which includes the removal of dilutant gases and other contaminants.
- Now, this biomethane gas is ready for further use in transportation and supplied to natural gas consumers.

After digestion, they screened and dewatered, producing a solid digested cake with total solids of 20-25%. As per the data, approximately 35 tonnes of digestate are produced /day. This material is used on-site by the Urban Growers Collective for many sustainable purposes, including urban agriculture. The liquid generated during dewatering is discharged into the municipal sewer system. Future nutrient recovery options are being explored, as this water is rich in nutrients and can serve as fertilizer.

The anaerobic digester at the Green Era Campus prevents the release of 42,500 tons of CO2-equivalent emissions by redirecting food waste from landfills. There are further plans to reduce more by capturing CO2 and converting it into food-grade products.

The project provides significant community benefits: a 7-acre farm produces 125 varieties of crops, supports training programs, and improves access to fresh food. There is one community education center, a retail store, and many other things that support education, local economic-social development

6. Japan Case Study

Source: (United Nations Environment Programme & International Solid Waste Association, 2024)

Economic growth does not have to go hand in hand with waste generation

Kitakyushu city in Japan sets an amazing example for the rest of the world by producing less than 0.42 kg/day of waste per person, even lower than sub-Saharan Africa's 0.46 kg/day. As this city is industrial and highly polluted, they focused on an environmentally

friendly approach rather than waste disposal. The process they adopted involves home-based segregation, followed by a strong emphasis on household composting. Additionally, they focus more on recycling and actively involve citizens in the waste management system.

To motivate and involve citizens, the government offers incentives and charges fees based on household waste volume rather than flat fees. Through these initiatives, Kitakyushu is gradually transforming into an eco-town, where it recycles materials from waste like household appliances and cars, while simultaneously raising environmental awareness among the people. This city experience provides valuable lessons to the world on waste management.

Citizens' participation is the key to its success. Support for government-like incentives drives the programme, aligns it, and transforms the waste management system, ultimately improving the city's overall ecosystem.

7. Chile Case Study

Biowaste Recycling and End-Market Development

In Chile, approximately 58% municipal waste is organic and is a major contributor to greenhouse gas emissions when disposed of in landfills. To overcome this environmental problem, a collaborative program, The Reciclo Orgánicos, has been established by the Ministry of the Environment and Climate Change of Canada, the Environment Ministry of Chile, and executed by Arcadis Canada, with support from Arcadis Chile and ImplementaSur. The programme focuses on reducing the harmful greenhouse gases emitted by existing landfills by capturing and utilizing them at the disposal site. Another focus is to prevent organic waste from going into landfills by using alternatives like composting and anaerobic digestion instead of disposal. The project summary is provided in Table 5.

Table 5: Chile project summary regarding organic waste management (The World Bank,2021).

Category	Challenges (Before Program)	Solutions (Program Actions)	Impacts (Results Achieved)
Greenhouse Gas Emissions	Landfills emitted around 4 million metric tons of GHG annually, mainly methane	Implementation of composting facilities, anaerobic digestors, and landfill gas capture systems	Over 6.65 megatons of CO_2 equivalent were reduced in the first 3 years

Public Health & Environment	Pollution threatens air quality and public health	Cleaner waste processing technologies and landfill gas-to-energy projects	Cleaner air, safer communities, and more sustainable waste services
International Commitments	Risk of failing Paris Agreement climate targets	Partnership with Canada and alignment with Canada's $2.65 billion global climate pledge	Stronger climate leadership and improved international credibility
Technology & Innovation	Limited infrastructure for organic waste recycling	Use of GORE® Cover Technology and blockchain-based MRV systems	Transparent monitoring and advanced waste management practices
Community Engagement	Low public awareness and limited municipal capacity	Training for municipal staff, awareness campaigns, and citizen participation	Support to 30+ communities and stronger local ownership
Energy & Economy	Landfills waste valuable energy potential	2 MW landfill gas power plant in San Felipe	Cheaper energy, new investments, and economic benefits

8. Singapore Case Study

Source: Singapore National Water Agency

Integrated Liquid Waste Management System

Singapore is a small island nation situated in South Asia with a population of more than 5 million. The daily water demand is 440 million gallons, and projections indicate it will double by 2065. As the economy and various industries expand, the pressure on water management intensifies.

To manage these challenges in water demand, Singapore has implemented an approach that develops an integrated system to collect every drop of water, reuse it, and desalinate seawater.

The city-state employs advanced technologies, including rainwater harvesting, wastewater recycling through its NEWater initiatives, and large-scale seawater desalination. This comprehensive strategy not only ensures a robust and sustainable water supply for the future, but it also states Singapore's efforts and commitments to manage the increasing water demand.

Singapore has a diversified water supply system: The "Four National Taps."

8.1. Local Catchment Water:

In this system, rainwater is collected through an extensive network of rivers, canals, and drains, and stored in 17 reservoirs, including the Marina Reservoir located in the city. To ensure that the waterways remain free from pollution, there are two separate collection systems in place: one for rainwater, which is stored in reservoirs before undergoing treatment for drinking purposes, and another for wastewater, which is collected through underground sewers and directed to a reclamation plant. Both systems work together to ensure that the water remains pollution-free.

8.2. Imported Water:

Singapore is allowed to take up to 250 million gallons of water per day from the Johor River under the 1962 Water Agreement.

8.3. NEWater:

This system includes a recycling process that treats used water into ultra-clean, high-quality water, supporting water demand-supply and accelerating Singapore's progress towards sustainability. This system uses membrane technology, and the water produced is called NEWater, which is internationally recognized by the WHO and USEPA as safe for drinking. To date, there are 4 NEWater Plants in Singapore.

(a) The major steps and technology in this process are:

- Collection of used water: Water is collected from industries and houses and sent to the reclamation plant. Then, in the plant, water is cleaned to meet standards, removing dirt, waste, and harmful substances.
- Microfiltration/Ultrafiltration: After initial treatment, the next and important stage is this process, in which treated water is filtered through membranes designed to remove microscopic particles.
- Here in Singapore, membrane bioreactor technology (MBR) is employed for water reclamation, effectively enhancing the purity, safety, and availability of the water.
- Reverse Osmosis: Reverse osmosis is a process that uses a semi-permeable membrane. The pore size of this membrane is very small, allowing only water molecules to pass through. Contaminants such as viruses, bacteria, heavy metals, and other pollutants cannot permeate the membrane due to its pore size limitations and are left behind.

- Ultraviolet Disinfection: Another important step in the NEWater process is ultraviolet disinfection, which kills any remaining traces of bacteria and viruses. Although the RO process already cleaned the water, as an extra measure for purity, this disinfection step is added. :The flowchart of the process is shown in Figure 6.

- The ultimate use of NEWater is for industrial purposes, which is transferred through a proper pipeline network. Another important use is that during the dry season, they refill their reservoirs with this NEWater, which is mixed with raw water, and then, after waterworks treatment, supplies people as tap water.

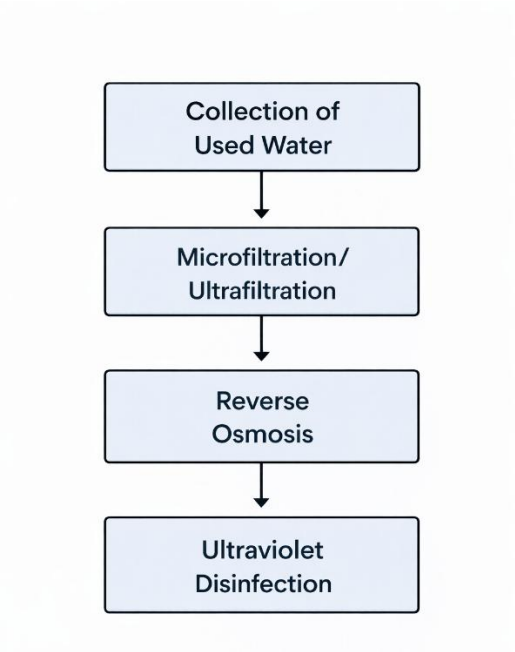


Figure 6: Steps under the NEWater treatment process

(b) Desalinated Water:

Desalination is an energy-intensive process that involves pushing seawater through special filters/membranes to remove salt and minerals, purifying the water. Here in Singapore, they use reverse osmosis. They are also focusing on another method, as it is an energy-intensive process, to ensure desalinated water remains sustainable in the future. There are 5 desalination plants operational in Singapore. A detailed diagrammatic presentation of the desalination process is shown in Figure 7.

The desalination process includes the following steps:

- Sea water flows through an open-sea network structure equipped with a mechanical rake course structure. Then chlorine is added to inhibit marine life throughout the entire system.
- The water subsequently enters the floating unit, where various chemicals are added to aggregate impurities, and air is pumped to allow small particles to float for removal.
- Next, the water flows through the gravity sand filtration unit, where heavy particles are filtered.
- From here, the pretreated seawater is passed through the reverse osmosis unit, where separation of dissolved solids occurs under high pressure through a semi-permeable membrane.
- Finally, this treated water undergoes further disinfection, and essential minerals are added to ensure it becomes potable.



Figure 7: Detailed diagrammatic presentation of the desalination process.

The dynamic conclusion of all the stages is ultimately Singapore's Water Loop, which includes:

- Collection of rainwater in stormwater drains, and then this rainwater is stored in reservoirs.

- After treatment at water treatment plants, the water is sent to homes and industries through pipeline networks for use.
- After that, the used water, collected simultaneously from homes and industries, flows into sewers and Deep Tunnel Sewerage Systems, then is sent to the reclamation water plant for cleaning.
- From here, some of the water is released into the sea, and some cleaned water is sent to NEWater plants for industrial use.
- Another process that continues alongside all these is seawater desalination in desalination plants.
- The entire movement of water is through the big pipeline networks.

Numerous case studies in liquid waste management share the common goals of ensuring safe water access for all and addressing the daily rise in wastewater generation, which poses significant environmental and health risks. Cities and countries face several challenges, including managing greenhouse gas emissions, controlling odors from open treatment facilities, and coping with water scarcity.

One notable approach is the use of anaerobic ponds in Santa Cruz de la Sierra, where covered anaerobic pond systems have been established. These systems utilize floating membranes to capture biogas generated during the anaerobic digestion of organic waste. This project is financially supported by the World Bank Group.

Another example is the case study of Lingyuan City, China, which has adopted a strategy focused on the supply, use, recovery, and restoration of water instead of merely discharging it. The key actions involve collecting wastewater and implementing tertiary treatment processes to enhance water quality for reuse. This treated wastewater is supplied to industries and returned to the ecosystem to restore the urban lake and its surrounding biodiversity and ensure that waste is treated as a resource. This initiative is also supported by the World Bank Group.

9. Colombia Case Study

Source: (United Nations Environment Programme & International Solid Waste Association, 2024)

Medellín and the Metropolitan Area of the Aburrá Valley, a model of corporate governance for public utility provision of sanitation service

Columbia municipalities are compelled to ensure that the population has access to water and sanitation within the urban perimeter. For this, there is a commercial enterprise, Empresas Públicas de Medellín (EPM), that provides water, sewerage, electricity, and gas services. Approximately 84 percent of collected wastewater is treated (secondary treatment) and released into the river.

The most important steps toward this success are long-term planning, investments, and financial support. In metropolitan areas, 98% of people have water connections, and 96% have sewer connections. The key points from the success story of the management of wastewater and sanitation are:

This company is owned by the municipality and managed professionally, with clear financial rules, proper professional management, and defined performance targets. The finances are kept separately from the municipal budget. There is a compulsion to develop a long-term plan (10 years) focused on expanding sewage and wastewater treatment, infrastructure planning, and financial arrangements.

If EPM cannot complete the project with its own resources, it seeks assistance from the local and national governments, international financial institutions, and others. This approach ensures the continuity of projects even in the event of a change in political leadership. Additionally, EPM also focuses on multiple funding sources, including user tariffs, the national government, and dedicated funds for water and sanitation networks.

Colombia has implemented a cross-hybrid system with tariffs differentiated by income group. In this system, high-income households pay higher tariffs, and low-income groups receive subsidies. They also invested in wastewater treatment infrastructure plants. More than 80 % water undergoes secondary treatment and is then discharged into the river. These steps lead to protecting the entire ecosystem. Apart from this, countries like Indonesia have wastewater treatment technologies, such as stabilization ponds, anaerobic facultative ponds, and maturation ponds. In Bangladesh, Dhaka city has a primary sediment tank, facultative lagoons, and chlorination.

10. The Best Practices for Solid-Liquid Waste Management

This report highlights best practices for solid-liquid waste management in medium- and large-sized urban areas in developing nations, where significant challenges are encountered. As time passes, these challenges will grow, so decision-makers should focus

on them and take important, effective action. Global best practices in solid and liquid waste management (SLWM) increasingly emphasize the circular economy, viewing waste as a resource to be reused rather than simply discarded. Prominent strategies adopted worldwide include mandatory source separation, advanced recycling technologies, Waste-to-Energy (WTE) systems, and strict regulations on hazardous substances.

According to a case study on Ireland, processing 235,000 tons of municipal waste at a waste-to-energy plant generates 18 MW of electricity. This represents a significant advancement in solid-liquid waste management. The waste streams include several types of solid and liquid waste, achieving a recovery rate of over 90%.

Building on this study, we should also focus on introducing similar waste-to-energy plants in the major cities along the Narmada River, especially since a plant in Jabalpur has already been established and is operating successfully.

Technology: Waste to Energy Plant

Another important takeaway is the need to store renewable energy in batteries for future use, promote the use of electric vehicles with on-site charging, and support ecological restoration by creating ponds, wetlands, and biodiversity.

The next analysis is from the US case study, centered on an Anaerobic digester that processes food waste collected from various sources and reduces reliance on landfilling.

Another important point to consider is that after food processing, a solid digested cake is produced, which is further used for many sustainable purposes. Furthermore, the liquid discharge produced during washing is a rich nutrient source and can be used as fertilizer.

Such systems bring communities together, support them in growing a variety of crops, offer training programs, and provide fresh food to society.

A key takeaway from these types of plants is that food waste can be used as a resource to generate fresh food In the Narmada Basin, such plants should be established to further sustainable processes.

Technology: Anaerobic Digester

Kitakyushu City in Japan sets a world-leading example in solid waste management by emphasizing household segregation and composting, recycling efforts, and active citizen involvement in the waste management process. The city implements a fee system that

charges residents based on the volume of waste generated, while also providing incentives to encourage waste reduction.

A key takeaway is that government involvement should align with citizen engagement, offering incentives to motivate residents to generate less waste.

Technology: Segregation, composting, and recycling

Another case study is from Chile, which focuses on reducing greenhouse gas emissions from organic waste in landfills. To this end, they want to reduce organic waste disposal and capture greenhouse gases. To this end, they implemented composting facilities, anaerobic digestors, and landfill gas capture systems. Here, they have a 2 MW landfill gas power plant that converts landfill waste into valuable energy.

Key takeaway: We should also focus on cities such as Jabalpur, Narmadapuram, Mandla, and East and West Nimar for organic waste management.

Technology: Composting. Anaerobic digester and landfill gas capture system

One of the recent and successful advanced case studies is Singapore's wastewater management. They created the water-loop system, in which not a single drop is wasted. Rainwater harvesting (storage in reservoirs), wastewater collection, treatment, and desalination are the key points from this case study. In this case, different types of technologies have been used.

Key takeaway: According to the study, there should be additional reservoirs in the Narmada Basin for rainwater harvesting, along with a water treatment plant to support water supply for commercial, domestic, and agricultural purposes. Another crucial point to be considered is water recycling, i.e., the collection and treatment of used water for reuse.

Technology: Micro/ultrafiltration, Reverse Osmosis, Ultra-violet disinfection, reservoirs, big pipeline networks for water movement.

Cities like Santa Cruz use anaerobic pond systems covered with floating membranes that capture biogas produced during the anaerobic digestion of organic waste. This is also an option to recycle organic waste into energy.

Another analysis is from Lingyuan city, which also focuses on wastewater recycling, with industrial wastewater returned to ecosystems. They believe waste should be treated as a resource.

Technology: Anaerobic ponds covered with floating membranes and tertiary wastewater treatment.

Another study on Colombian water and sanitation management found that effective government initiatives, budget, and public involvement can lead to effective waste management.

Key takeaway: We should prioritize long-term waste management plans and invest in treatment infrastructure for both solid and liquid waste. There should be multiple funding sources to ensure the project's successful completion. User tariffs should also be based on income groups.

Key technologies: Secondary treatment

11. Global Insights and Recommendations for Sustainable Waste Management

The integration of these technologies into solid-liquid waste management in the Narmada Basin is poised to deliver significant environmental and socio-economic benefits. Transforming waste into energy resources, such as batteries, will be most effective when demand is high or during periods when solar or wind energy sources are unavailable. Reducing greenhouse gas emissions, transitioning from disposal to processing, and reducing organic waste can protect the river and its surrounding ecosystems, ultimately enhancing overall water quality. Improvements in the management of organic and other waste increase the likelihood of reviving aquatic life, including the reemergence of sensitive species, while also improving water quality, which ensures a safe and reliable water supply for the community.

In addition to these efforts, fostering public engagement, promoting government initiatives, implementing training programs, and supporting river conservation will lay a robust foundation for a sustainable river management system in the long term.

Policy and Governance: Policymakers should develop policies focused on the waste hierarchy for solid-liquid waste management, i.e., prevention, reuse, recycling, recovery, and disposal, for all cities There should be serious acceptance of landfills as a no-option or last option for waste disposal.

Policymaking also ensures that product designs, manufacturing, and other packaging targets prioritize eco-friendliness and recyclability, and that products provide consumers with detailed information.

There should also be a fee system: the more waste households generate, the higher the fees; for less waste, incentives for citizens.

Technology and Infrastructure: There must be waste-to-energy plants to process non-recyclable materials, and they can store energy in batteries. Given the significant greenhouse gas emissions from organic waste, anaerobic systems must be implemented in cities. Installation of a landfill gas capture system covering anaerobic ponds, good drainage systems, and a circular loop for water recycling; sanitary landfills construction, all these technologies and infrastructure will help achieve the sustainability goal and support a healthy ecosystem.

Finance and Markets: In addition to local governments, proper funds include user tariffs, national funds, climate funds, and development funds.

Community and Health: There should be clear guidelines and strict rules for doorstep waste collection. Before collecting waste, it should be properly segregated, and for that, citizens should get incentives.

More focus should also be placed on proper training for citizens on sanitation, waste management, and health risks.

12. Current Scenario of Solid-Liquid Waste Management in India

India is facing rapidly escalating waste management challenges due to urbanization and population growth, generating approximately 62 million tonnes of municipal solid waste (MSW) annually, which is projected to reach 165 million tonnes by 2030. In parallel, the country produces around 72,368 million litres per day (MLD) of sewage, with significant treatment gaps that continue to hinder environmental sustainability. Over recent years, MSW generation has increased from about 62 million tonnes in 2021 to nearly 67 million tonnes by 2026, driven by an expanding urban population expected to reach 35–53% by 2050. Daily waste generation has risen to nearly 185,000 tonnes, including substantial contributions from plastics (5.6 million tonnes), e-waste (1.5 million tonnes), and hazardous waste (7.9 million tonnes). Although solid waste processing efficiency has improved considerably from 18% in 2014 to around 54% in 2026, due to widespread adoption of composting, biomethanation in multiple states, and waste-to-energy plants in selected regions, significant gaps remain, with nearly 30% of waste still unaccounted for or ending up in landfills. On the liquid waste front, only about 31,841 MLD (44%) of the total sewage generated is treated, with actual utilization of installed capacity ranging between 75–80%, leaving approximately 35,700 MLD untreated and contributing to severe river pollution. Major metropolitan cities such as Delhi and Mumbai lead in treatment capacity, with 2,330 MLD and 2,130 MLD, respectively, yet still fall short of total sewage generation. A variety of treatment technologies are employed across the country, including UASB followed by polishing units, Sequential Batch Reactors (SBR), Cyclic Activated Sludge Technology (CTECH), Moving Bed Biofilm Reactors (MBBR), Fluidized Aerobic Beds (FAB), and Activated Sludge Processes (ASP). Emerging solutions such as modular treatment units for faecal sludge, constructed wetlands for decentralized applications, and advanced oxidation processes, such as Fenton treatment, are also gaining traction, along with an increasing emphasis on biogas recovery in advanced sewage treatment plants.

13. Impact of Poor Waste Management in India

India generates approximately 1.85 lakh tonnes (185,000 tonnes) of municipal solid waste (MSW) daily, amounting to over 67 million tonnes annually, with nearly 70% of it mismanaged through open dumping or unlined landfills, thereby exacerbating public health crises, environmental degradation, and socioeconomic burdens in urban and peri-

urban areas **(Ravish, 2025)**. These 2,452 largely non-engineered sites account for 15% of national methane emissions (1.25–1.68 Tg/year), far exceeding models, as satellite data from Delhi's Ghazipur and Bhalswa landfills show emissions are double official estimates **(Mor et al., 2024)**. Groundwater contamination is rampant, with leachate pushing nitrates in 440 districts (56% of India) above 45 mg/L safe limits and chlorides to 4,000 mg/L near Bhalswa, alongside heavy metals like Cd and Cr **(Ahamad et al., 2019)**. This mismanagement undermines climate goals, biodiversity, and water security, disproportionately impacting 1.5–4 million informal waste workers, mostly women, migrants, and minorities who recover 20-30% recyclables but earn under $4/day amid hazardous conditions **(Ferronato & Torretta, 2019).**

13.1. Human Health/Disease Transmission

Uncontrolled dumpsites like Ghazipur (Delhi) and Deonar (Mumbai) breed vectors, spiking dengue, malaria, and cholera. CPCB links these to 62 million tonnes MSW/year, with leachate spreading E. coli and typhoid in adjacent slums **(Ravish, 2025)**. Waste workers show 21-47% respiratory ailments and 73% injuries per Karnataka studies, while burning releases dioxins, causing asthma and cancer. **(Alam et al., 2024).**

13.2. Ecological Disturbance

Leachate from non-engineered sites contaminates groundwater, including Bandhwari (BOD 2,216 mg/L, COD 7,998 mg/L), killing aquatic life and bioaccumulating metals such as Cd and Cr in soils. **(Srivastava et al., 2023)**. Rivers receive 1-3 billion microplastics daily from MSW, disrupting Ganga ecosystems and fisheries; landfills fragment habitats and extinguish insects at Kodungaiyur. **(Lechthaler et al., 2021)**.

13.3. Socioeconomic Decline

Inefficient management forfeits ₹51,000 crore in e-waste recovery and raises remediation costs (₹2,000-3,000/kL leachate), per NITI Aayog. Waste pickers (51% lower middle class) endure hazards without support. Landfills depress property values by 5.5-7.3% (Pune), inflate healthcare costs by 15-25% near sites, and perpetuate slum poverty. **(Ravish, 2025)**.

Poor waste management in India drives socioeconomic decline through lost economic opportunities, degraded livelihoods, health costs, and property devaluation. Government

reports like CPCB and NITI Aayog, alongside peer-reviewed studies, quantify burdens on informal workers, low-income communities, and urban areas.

Informal Sector Hardships

Waste pickers, vital to 20-30% recycling, endure hazardous conditions without protection; a Ghaziabad study of 200 pickers found 51% were lower middle class and 42% were upper lower class, mostly illiterate and underprivileged. CPCB notes they face injuries and exclusion, limiting income despite recovering valuables worth thousands of crores.

Economic Losses

NITI Aayog reports that inefficient scrap management results in thousands of crores in recycling revenue forgone, e.g., ₹7,500 crore annually from waste tires due to a lack of standards; e-waste extractables alone at ₹51,000 crore, 60% recoverable. Remediation costs for landfills are ₹2,000-3,000/kL for leachate and ₹5-10 crores/hectare for reclamation.

Health and Productivity Costs

Landfill-adjacent clinics report 15-25% higher healthcare expenses due to chronic illnesses and outbreaks; Kodungaiyur and Deonar surges during monsoons are linked to pollution. This burdens low-income residents, reducing productivity and perpetuating poverty cycles.

Property and Urban Impacts

Proximity to landfills depresses residential values by 5.5-7.3%, vanishing beyond 10 km, per Pune studies; developers add amenities to offset. Marginalized groups near sites face land devaluation and exclusion.

14. Solid- Liquid Waste Management in different cities in India

14.1. Solid Waste Management

Solid and liquid waste management in Indian cities varies widely, with top performers like Indore and Surat excelling in segregation, while megacities like Delhi and Mumbai lag in processing capacities. CPCB and NITI Aayog reports highlight gaps in treatment: urban

MSW is processed at 156,000 tonnes/day (54%), and sewage generation is 72,368 MLD, against a capacity of 31,841 MLD.

Solid waste management (SWM) in India represents a dynamic yet challenging sector characterized by increasing waste generation, uneven infrastructure development, gaps in policy enforcement, and significant socio-environmental impacts. As reported by the Central Pollution Control Board (CPCB), the country generates approximately 1,70,339 tonnes per day (TPD) of municipal solid waste (MSW), with an average collection efficiency of around 92%. Despite this, only about 54% of the collected waste undergoes scientific processing, while 24% is disposed of in landfills, and the remaining 22% is either untreated or improperly managed **(Ravish, 2025)**.

Per capita waste generation exhibits substantial regional variation as shown in Figure 8, ranging from as low as 21 g/capita/day in Meghalaya to over 500 g/capita/day in Delhi, reflecting differences in consumption behavior and waste management infrastructure. Furthermore, stark disparities exist in treatment efficiency across regions: states such as Chhattisgarh, Lakshadweep, and the Andaman & Nicobar Islands report efficiencies exceeding 90%, whereas regions like Arunachal Pradesh and Puducherry demonstrate significantly lower performance, with treatment rates below 15%. These highlights pronounced regional inequalities in the effectiveness of SWM systems across India.**(Ravish, 2025)**.

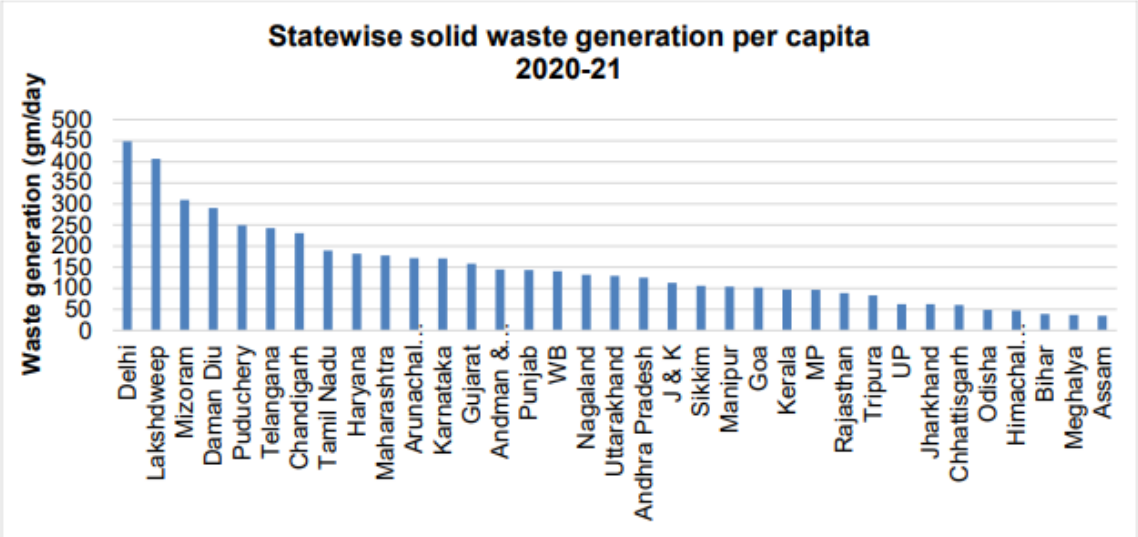


Figure 8: State-wise per capita solid waste generation

(Source: Annual Report on Solid Waste Management (2020-21)

Indore leads with 100% door-to-door collection and source segregation, processing 1,500+ tonnes/day via bio-methanation and composting; Pune and Bhopal treat 91,000 tonnes/day across the nation in model cities. Delhi generates 10,990 TPD of MSW (49.65% processed), Mumbai generates 9,400 TPD (52% processed), Bengaluru focuses on 100% efficiency via dry waste parks. Surat and Rajkot achieve high recycling via PPP models.

14.2. Trend in solid waste processing:

It is observed that the maximum percentage of solid waste treated is in Chhattisgarh (100 %), followed by DDDNH (88.76), Goa (87.04 %), Andaman and Nicobar Islands (84.7 %) &Madhya Pradesh (80.67 %) in that order. Figure 9 shows the percentage of state-wise solid waste treatment.

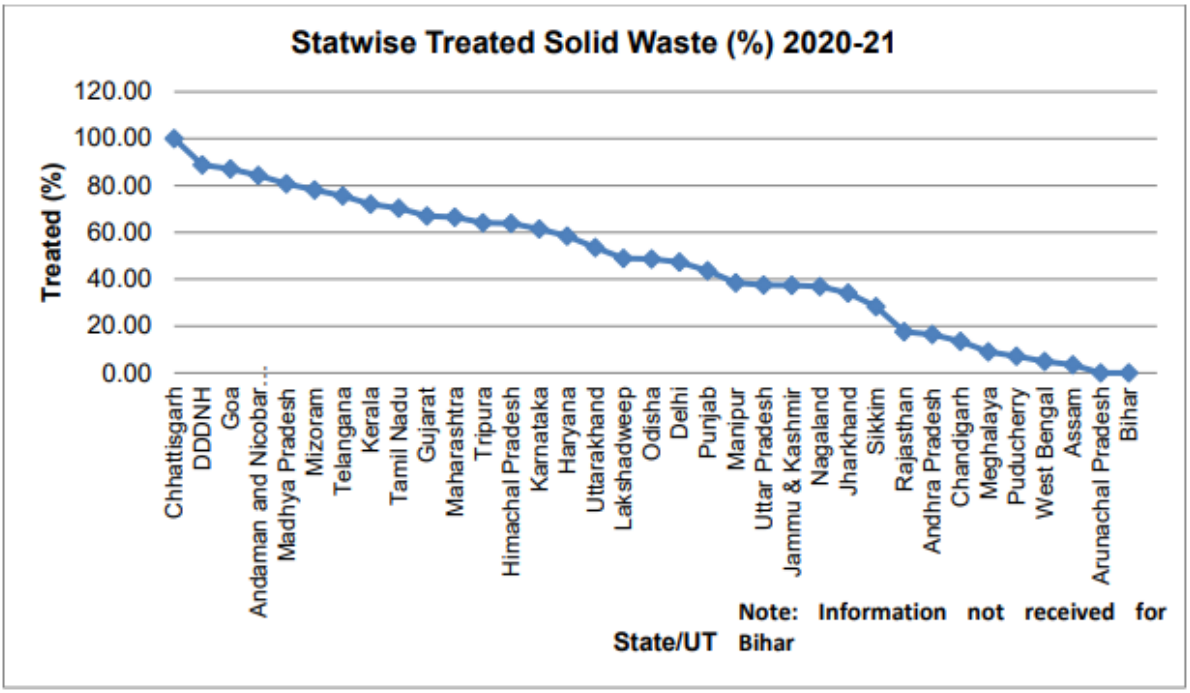


Figure 9: Percentage of state-wise solid waste treatment

14.3. Liquid Waste (Sewage) Treatment

The estimated wastewater generation in rural areas is around 39,604 million litres per day (MLD), whereas urban areas generated approximately 72,368 MLD during 2020–21. Urban wastewater volumes are nearly double those in rural regions, primarily due to higher

water use for flushing and the presence of organized sewage systems. This greater availability of water for sanitation has contributed to improved living standards in cities. Consequently, rapid urbanization and increased migration toward urban centers have intensified the need for effective wastewater management, making it a matter requiring urgent attention.

In the Indian context, data on wastewater generation, as shown in Figure 10, collection, treatment, and reuse remain fragmented, inconsistently monitored, and often unavailable across regions. Such information is vital for achieving sustainable development and meeting the United Nations' Sustainable Development Goals (SDGs). SDG 6.3 focuses on reducing untreated wastewater and enhancing reuse, while linking with sanitation (6.a), clean energy (7.a), sustainable urbanization (11.3), waste reduction (12.5), and climate action (13.2). Therefore, improving wastewater data systems and reuse practices is essential for India's sustainable growth.

As urban populations grow, water demand across sectors—including irrigation, domestic use, industry, and energy is intensifying. By 2050, India's total water requirement is projected to reach 1,447 km^3, with 74% allocated to irrigation and the remaining distributed among drinking water, industry, energy, and other uses. However, increasing urban demand, often prioritized over rural needs, is driving greater water transfers from rural and peri-urban areas to cities. Many expanding cities are located within major river basins, extracting freshwater and discharging untreated or partially treated wastewater back into these systems, thereby degrading water quality and creating significant challenges for both urban and rural water management.

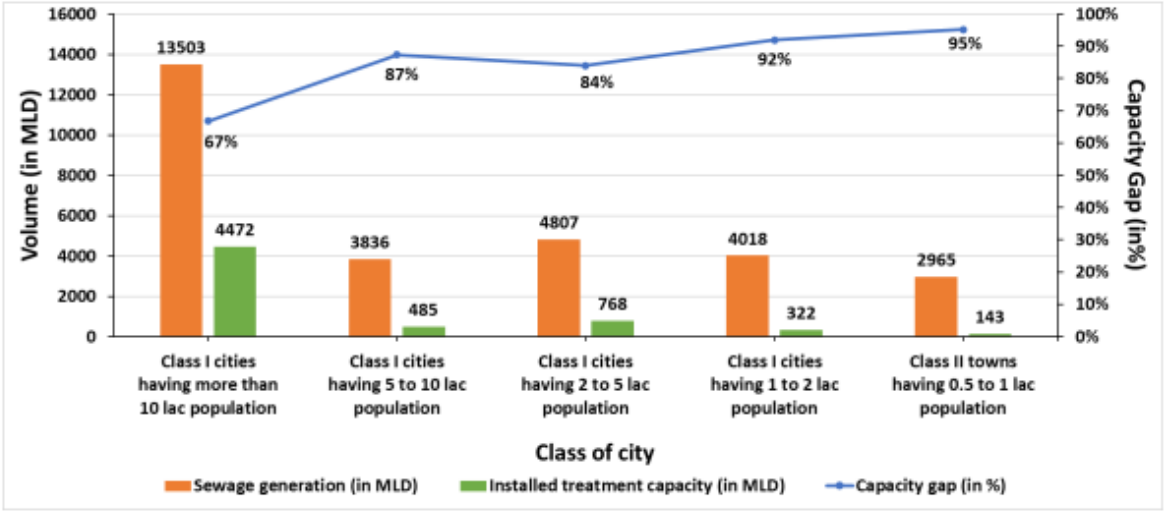


Figure 10: Wastewater generation and treatment capacity gap at the city level in India (CPCB, 2022)

Delhi has the highest capacity at 2,330 MLD (29% of the metro total), followed by Mumbai at 2,130 MLD (26%). National utilization is 75-80%, with a gap of 35,700 MLD of untreated water. Chennai and Bengaluru treat approx.50% of the sewage generated; Indore's 825 MLD plant emphasizes reuse for irrigation. Class-I cities collectively handle 11,554 MLD (32% of generation). Table 6 presents an assessment of Solid and Liquid Waste Management Infrastructure and Performance Across Selected Indian Cities.

Solid and Liquid Waste Management Infrastructure Comparisons

Table 6: Assessment of Solid and Liquid Waste Management Infrastructure and Performance Across Selected Indian Cities

City	MSW Generation (TPD)	MSW Processing (%)	Sewage Generation (MLD)	STP Capacity (MLD)	Source
Delhi	10,990	49.65	~3,000	2,330	CPCB/NITI
Mumbai	9,400	52	~2,800	2,130	CPCB/NITI
Bengaluru	6,500	Improving to 100	~1,900	~1,000	CEEW/NITI

City	MSW Generation (TPD)	MSW Processing (%)	Sewage Generation (MLD)	STP Capacity (MLD)	Source
Indore	1,500+	100	825	825	Smart City/CPCB
Surat	~2,500	High (model)	~600	~500	CEEW/NITI

15. Solid- Liquid Waste Management in different cities in India case study

Solid-liquid waste management case studies in Indian cities reveal stark contrasts: Indore serves as a national model for integrated solid waste management, while megacities like Mumbai and Delhi showcase persistent challenges in both solid and liquid streams. Research and government evaluations highlight scalable lessons from segregation drives and STP upgrades.

15.1. Indore: Solid Waste Success

Indore generates more than 1,115 tonnes/day MSW, achieving 100% door-to-door collection and source segregation since 2016 via pilot expansions and citizen incentives. Wet waste undergoes bio gasification and composting; dry waste goes to bio-CNG plants (the largest in India, cutting 130,000 tons CO2e/year). Engineered landfills and bioremediation of legacy dumps ensure zero open dumping, earning it the title of India's cleanest city repeatedly.

15.2. Mumbai: Mixed Challenges

Mumbai handles 9,400 TPD MSW with 52% processing via landfills like Deonar (overloaded at 9,000 TPD) and emerging waste-to-energy plants; segregation lags at 20-30%. Liquid waste has a 2,130 MLD STP capacity, against a 2,800 MLD generation rate, with a 26% national share, but frequent overflows pollute coastal ecosystems during monsoons.

15.3. Delhi: Capacity Gaps

Delhi processes 49.65% of 10,990 TPD MSW through Okhla plants and bio methanation, but Ghazipur legacy dumps persist. Sewage treatment at 2,330 MLD covers approx. 29% of the 3,000 MLD load is untreated and enters the Yamuna; upgrades focus on reuse, but

utilization hovers at 75%. Table 7 shows Solis waste key practice, liquid waste STP utilization, and outcomes and challenges in the following cities.

Table 7: Comparative Analysis of Solid and Liquid Waste Management Practices and Performance in Major Indian Cities

City	Solid Waste Key Practice	Liquid Waste STP Utilization	Outcomes/Challenges	Source
Indore	100% segregation, bio-CNG	825 MLD (full)	Model city, methane reduction	smartcityindore.org/solid waste
Mumbai	Partial WTE, landfills	2,130 MLD (approx.76%)	Floods, marine pollution	tandfonline.com/ **(Dutta & Jinsart, 2020)**
Delhi	Bio-methanation (50%)	2,330 MLD (approx.78%)	Yamuna pollution, overloads	mospi.gov.in
Bengaluru	Dry waste parks, 100% goal	approx.1,000 MLD (approx.53%)	Tech-driven but scaling issues	niti.gov.in/Waste-Wise-Cities

16. Waste Framework Directive in India

India lacks a direct equivalent to the EU's Waste Framework Directive but has evolved comprehensive national regulations under the Solid Waste Management (SWM) Rules, first notified in 2016 and superseded by the SWM Rules, 2026, effective April 1, 2026. These rules, issued by the Ministry of Environment, Forest, and Climate Change under the Environment (Protection) Act, 1986, promote a circular economy, extended producer responsibility, and sustainable waste handling to address the 62+ million tonnes of annual MSW.

16.1. Key Provisions (SWM Rules 2016)

Waste generators must segregate into three streams: wet (biodegradable), dry (recyclable), and domestic hazardous (e.g., diapers, cleaners), with local bodies ensuring door-to-door collection and user fees. Urban local bodies with 1 million+ population must establish processing facilities within two years, including composting, bio methanation, and waste-to-energy; legacy dumps require biomining. Bulk generators (e.g., hotels, malls) handle on-site processing, while states identify landfill sites per strict criteria (e.g., 500m buffer from habitations).

16.2. Updates in SWM Rules 2026

Four-stream segregation is now mandatory: wet, dry, sanitary, and exceptional care waste, with Extended Bulk Waste Generator Responsibility (EBWGR) certificates for off-site processing. Material Recovery Facilities (MRFs) are formalized for sorting, including e-waste; online portals enable tracking, faster land allocation, and quarterly waste remediation reports. Hilly/island areas receive tailored guidelines that emphasize polluter-pays environmental compensation.

16.3. Implementation Framework

CPCB monitors compliance, with states forming policies and panchayats integrating rural plans; waste pickers gain registration. Extended Producer Responsibility targets plastics and aligns with global standards, though challenges persist in enforcement across 7,000+ urban bodies.

17. Different state case study

India's state-level solid and liquid waste management showcases diverse approaches, with standout case studies from Madhya Pradesh (Indore model), Gujarat (Surat), Maharashtra (Pune), Chhattisgarh (Raipur), and Rajasthan (Jaipur). These draw on government initiatives such as the Swachh Bharat Mission and the SWM Rules 2016/2026, emphasizing segregation, decentralization, and PPPs.

17.1. Madhya Pradesh: Indore Model

Indore (3M+ population) achieves 100% door-to-door collection of 1,500 TPD MSW via GPS-tracked vehicles and three-stream segregation, processing wet waste through 40+ decentralized biomethanation plants and a 700 TPD bio-CNG facility. Legacy waste

bioremediation cleared more than 1M tonnes; liquid waste via 825 MLD STP supports irrigation reuse and earns national awards.

17.2. Gujarat: Surat Success

Surat (7M metro) manages 2,500 TPD MSW with 95% segregation, exporting recyclables via MRFs and composting 700 TPD organics; waste-to-energy pilots reduce landfill use. Sewage treatment at 500+ MLD (high utilization) integrates tertiary reuse, backed by strong municipal enforcement and citizen fines for non-compliance.

17.3. Maharashtra: Pune Approach

Pune (7M metro) processes 91% of 2,200 TPD MSW through dry waste parks, biomethanation, and a compressed biogas plant; decentralized composting in 26 wards cuts transport emissions. STP capacity (approx. 1,200 MLD) treats 70% of sewage, with challenges of slum integration addressed through NGO partnerships.

17.4. Chhattisgarh: Raipur Reforms

Raipur (2M+) introduced 100% door-to-door collection post-2016, with a 1,000 TPD integrated SWM plant at Sakri featuring secondary segregation, RDF production, and scientific landfilling under a PPP. Smart bins and bans on open burning improved compliance, despite initial court fines for site norms.

17.5. Rajasthan: Jaipur Challenges

Jaipur (4M metro) handles 2,500 TPD of MSW at Langadiyawas landfill, facing overload and leachate issues; biomining pilots and plasma gasification proposals aiming to achieve waste-to-energy. Sewage treatment capacity lags at approx.600 MLD, while higher generation is expected, with GIS mapping aiding route optimization. Table 8 defines SWM Strengths, liquid waste status, and challenges in Indian states and their key cities.

Table 8: State-wise Key City Performance in Solid and Liquid Waste Management – Strengths, Status, and Challenges in India.)

State	Key City	SWM Strengths	Liquid Waste Status	Challenges	Sources

Madhya Pradesh	Indore	100% segregation, bio-CNG	825 MLD STP is in full use	Scaling to rural areas	(Rasmeet, 2021)
Gujarat	Surat	MRFs, composting	High reuse	Industrial waste integration	https://www.ceew.in/
Maharashtra	Pune	Decentralized processing	70% STP coverage	Slum coverage	(Ghosh et al., 2025)
Chhattisgarh	Raipur	PPP plant, 100% collection	Emerging STPs	Legacy dumps	https://niua.in**ment.pdf**
Rajasthan	Jaipur	GIS routes, biomining	Capacity gaps	Landfill overload	

18. The Best Practices for Solid-Liquid Waste Management in India

India's best practices for solid-liquid waste management emphasize source segregation, decentralized processing, public participation, and circular-economy principles, as demonstrated by model cities and the updated SWM Rules 2026. These draw from high-performing cases like Indore and Surat, integrating technology and policy enforcement.

18.1. Source Segregation

Mandate four-stream segregation (wet, dry, sanitary, domestic hazardous) at households and bulk generators (>100 kg/day), enforced via user fees and fines; Indore achieves 100% compliance through awareness campaigns and GPS vehicle tracking, enabling 700 TPD bio-CNG from wet waste. Bulk generators must obtain EBWGR certificates for off-site processing.

18.2. Decentralized Processing

Compost or biomethane wet waste locally to cut transport (e.g., Pune's 26 ward plants process 300 TPD); Surat composts 700 TPD organics via PPPs. Dry waste routes to MRFs

for recycling (recovering 20-30% value), with RDF for waste-to-energy. Best practices and States of India are defined in Table 9.

Table 9: State-wise Best Practices in Solid and Liquid Waste Management Across India (https://mohua.gov.in/pdf/627b8318adf18Circular-Economy-in-waste-management

S. No.	State/UT	Best Practices
1	Andhra Pradesh	• CCTV surveillance at dumpsites in 70 ULBs (ongoing in 54 ULBs) • 99% households covered under door-to-door collection
2	Andaman & Nicobar Islands	• 100% segregated waste collection • Spot fines for littering • User fee collection for door-to-door services • Installation of water ATMs
3	Chandigarh	• Biomethanation plants for market waste (vegetable, meat, fish, etc.) • Onsite composting by bulk waste generators
4	Chhattisgarh	• 100% waste processing • MIS-based monitoring system • PIUs and PMUs established • SLRM centres in 164/166 ULBs • Waste picker registration and surveys • PPE provided to sanitation workers
5	Delhi	• 100% door-to-door MSW collection achieved
6	Goa	• 87% waste processing

		• Decentralized composting in housing societies, markets, and gardens
7	Gujarat	• 100% door-to-door MSW collection achieved
8	Haryana	• Cluster-based integrated SWM approach (14 clusters: WTE + Compost/RDF)
9	Himachal Pradesh	• Pit composting is operational in all ULBs
10	Jammu & Kashmir	• Composting using auto-composters and disintegrators
11	Kerala	• Multiple composting techniques: aero bins, bio bins, pipe composting, ring composting
12	Karnataka	• 100% door-to-door MSW collection achieved
13	Lakshadweep	• Distribution of blue & green bins for segregation • 455 sanitation workers engaged
14	Madhya Pradesh	• 100% door-to-door collection • Composting in 130 towns • RDF in 47 towns • Biogas in 3 towns • Vermicomposting in 33 towns • 50 dumpsites reclaimed • Promotion of home composting
15	Maharashtra	• “Harita Maha City Compost” brand developed • Web portal for SWM monitoring • Financial & technical support to ULBs • Model SWM tender documents • Integration of rag pickers into the formal system
16	Manipur	• Waste-to-energy plant (pyrolysis) under trial at Langdeng
17	Mizoram	• Training programs for waste workers • Use of eco-micro solutions for odor and flies control

18	Nagaland	• Bailing machines in 11 district HQs • Dumpsite development in Dimapur
19	Puducherry	• Authorization granted for SWM processing units • Land identified for SLF & MRF • Nisarguna technology (BARC) implementation • Pipe composting adopted
20	Sikkim	• GPS-enabled waste transportation
21	Tamil Nadu	• Barrel composting in 25 wards • Micro Compost Centres (MCC) with leachate recycling
22	Telangana	• 100% door-to-door collection • FSTPs in 2 ULBs • 67 ULBs grouped into 5 clusters for biomining (PPP mode)
23	Uttarakhand	• GPS-enabled waste collection vehicles • Onsite composting in all ULBs

19. Liquid Waste Reuse

Treat sewage to tertiary standards for irrigation/reuse; Indore's 825 MLD STP achieves 100% utilization, while Delhi upgrades focus on 2,330 MLD capacity with biogas recovery. Constructed wetlands and Fenton oxidation suits are suitable for small towns.

19.1. Learning from Case Studies: National Mission for Clean Ganga (NMCG), India

The National Mission for Clean Ganga (NMCG) under the Namami Gange Mission represents a comprehensive initiative for river rejuvenation, focusing on pollution control, ecological restoration, and sustainable development. The Ganga basin, covering about 26% of India’s land area and supporting nearly 43% of its population, has been central to these efforts.

Launched in 2014 with a strong financial commitment, the mission integrates multiple projects related to sewerage infrastructure, wastewater treatment, industrial pollution control, and riverfront development. A five-tier institutional framework—from national to district level—ensures coordinated governance and effective implementation.

Key interventions include large-scale sewage treatment infrastructure, decentralized systems such as faecal sludge and septage management (FSSM), and the adoption of innovative technologies such as bioremediation and nature-based solutions. Public-private partnership models such as the Hybrid Annuity Model (HAM) and "One City One Operator" have strengthened project execution and long-term sustainability.

The mission also emphasizes industrial effluent treatment through CETPs, strict monitoring of polluting industries, and adoption of cleaner production technologies. GIS-based tools, remote sensing, and geoportals (e.g., Bhuvan Ganga) support data-driven decision-making and basin-level planning.

Public participation remains a core component through awareness campaigns such as Ganga Utsav and Ganga Quest, as well as through community engagement via Ganga Praharis and local institutions.

Overall, NMCG demonstrates an integrated approach combining institutional reform, technological innovation, stakeholder participation, and policy support, serving as a model for sustainable urban wastewater management in India.

20. Tech and Monitoring

Deploy smart bins, GIS route optimization, and centralized portals (per 2026 rules) for real-time tracking; vacuum systems in commercial hubs reduce handling risks. Legacy dump bioremediation, as in Raipur, clears 1M+ tons via biomining. Best practices are shown for solid and liquid waste examples in Table 10.

Table 10: Comparative Overview of Key Practices in Solid and Liquid Waste Management with Representative Examples and Associated Benefits

https://www.niti.gov.in

Practice	Solid Waste Example	Liquid Waste Example	Benefits

Segregation	Indore (100%)	STP pre-treatment	High recovery rates.
Decentralization	Pune composting	Wetlands (rural)	Emission cuts.
Tech Integration	GPS/MRFs (Surat)	Biogas STPs (Delhi)	Compliance tracking
PPP Models	Raipur plant	Reuse irrigation	Cost efficiency.

21. Conclusion and Future Recommendation

India's solid-liquid waste management landscape encapsulates a complex interplay of escalating challenges and transformative successes, generating over 67 million tonnes of municipal solid waste (MSW) and 72,368 million liters per day (MLD) of sewage annually, with only 54% MSW processing and 44% sewage treatment capacity amid widespread ecological disruptions from leachate pollution, heavy metal soil contamination, and microplastic influx into rivers like the Ganga. Health crises proliferate near 2,452 non-engineered dumpsites, where vector-borne diseases and respiratory ailments afflict waste workers and slum residents, while socioeconomic declines manifest in forfeited recycling revenues exceeding ₹51,000 crore and property devaluation by 5-7% adjacent to landfills. Nevertheless, exemplary practices in Indore—achieving 100% door-to-door segregation and bio-CNG production—Surat's PPP-driven recycling, and Pune's decentralized composting illuminate viable pathways under the SWM Rules 2016/2026 framework, proving that integrated technology, public participation, and policy enforcement can minimize landfilling, recover resources, and align with circular economy goals, provided these models scale nationally to avert projected 165 million tonnes MSW by 2030

To bridge these gaps, policymakers must mandate uniform four-stream segregation (wet, dry, sanitary, hazardous) across all urban local bodies via SWM Rules 2026-compliant digital portals, user fees, and incentives, targeting 90% compliance within three years; simultaneously, replicate Indore's biomethanation and Pune's ward-level plants to decentralize 80% MSW processing, reducing transport emissions and legacy dumps. For liquid waste, upgrade STP capacity by 10,000 MLD emphasizing tertiary treatment, biogas recovery, and reuse for irrigation/industry, alongside constructed wetlands for smaller towns; integrate 1.5-4 million informal waste pickers through formal registration, PPE

provision, and fair wages to harness their 20-30% recycling contribution; and allocate 2-3% of municipal budgets to bioremediation while leveraging public-private partnerships and extended producer responsibility for plastics/e-waste, monitored through CPCB dashboards for accountability and nationwide transformation.

22. Declaration

AI declaration: An Artificial Intelligence tool was used solely to enhance language clarity and quality.

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